FEASIBILITY STUDY · TECHNICAL PACK · V1.1 · MAY 2026

KARNETH
STRATA

Karneth Strata

DOMAIN-AGNOSTIC COORDINATION PLATFORM

ENGINEERING EVIDENCE BEHIND THE FEASIBILITY CLAIM

The back tier of the Karneth Strata feasibility pack. Engineering evidence for the domain-agnostic coordination architecture, demonstrator state, sim-to-real validation methodology, a wildfire-mission worked example showing the same code serving civil missions, regulatory posture, technical roadmap, technical risk register, and a glossary. Designed for review by an investor's technical adviser.

FRONT TIER: KS_FEASIBILITY_SEED_V1.1 READ ALONGSIDE: KS INVESTOR BRIEF V1.1

CLASSIFICATION: SHAREABLE TECHNICAL PACK

CONTENTS

§1 · ARCHITECTURE & STACK

1 How the system is composed

Karneth Strata is built in four horizontal layers and three orthogonal cross-cuts. The horizontal layers concern data flow from sensor to operator decision; the cross-cuts concern properties the whole system must hold simultaneously (sovereignty, safety, auditability).

```
OPERATOR          [ Operator station, tactical map, mission planner ]
                                     |
DECISION          [ Mission planning, kill-chain, EMCON policy,
                    fire-request governance, evasion budget, NFZ ]
                                     |
COMMS             [ BURST MESH - transmitted diffs, resilient,
                    EMCON-aware, portable to real radios ]
                                     |
PLATFORM ADAPTER  [ PX4, MAVLink, PiDog, ground vehicles,
                    maritime (future) - JSON vehicle profiles ]
                                     v
HARDWARE          [ Air / ground / maritime - any aligned supplier ]

CROSS-CUTS (apply at every layer):
  -  SAFETY     altitude clamps, NFZ, evasion bounds, emergency
  -  SOVEREIGN  Lead-Box AI, in-house code, audited supply chain
  -  AUDIT      inspectable record of every decision, replayable
```

This separation is the architectural payoff of the team's discipline. Each layer has narrow, inspectable interfaces. Adding a new platform is a Platform Adapter exercise that does not perturb the Decision or Operator layers. Adding a new mission type is a Decision Layer exercise that does not perturb the Comms or Platform layers. Adding a new comms transport (e.g. a new radio) is a Comms Layer swap that does not perturb anything above it.

Domain-agnostic by construction

The same separation makes the architecture domain-agnostic. The Decision Layer does not know whether it is running a defence kill chain or a wildfire response chain — both express as a corroborate-authorise-engage-assess cycle over a platform-shared blackboard. The Comms Layer does not know whether silence is being enforced because of enemy SIGINT or simply not needed in a civil scenario; EMCON is a runtime preset, not a hard-coded posture. The Operator Layer does not know whether the operator is a military commander or a fire incident commander; the same surfaces serve both, with mission-string and authority-model presets configured per deployment.

The wildfire-mission worked example in §4 makes this concrete.

Key design choices that matter for feasibility

- **Simulator shares code with the real system.** The simulator is not a parallel prototype; it drives the same Decision and Comms layers used in live integration. This is what allows demonstrator confidence to transfer to fielded confidence — a property most defence-tech demos cannot claim.
- **Transmitted-diffs, not shared memory.** Cross-platform state propagates as serialised diffs over the BURST MESH queue. The architecture is portable to real radios because nothing relies on the simulator's shared memory; the wire format is the contract.
- **EMCON is enforced at the display level.** Operator displays read from a transmitted-only telemetry view, not from source values. Radio-silence discipline is a code property, not an operator-discipline expectation.
- **JSON-driven vehicle profiles.** Per-platform parameters (chassis dimensions, clearance, NavMesh radius, icon assets) live in a single configuration file, not hardcoded. Adding a new platform class is a configuration change, not a code change. The same pattern extends to mission-preset profiles (defence vs civil).
- **Centralised helpers, embedded-grade allocation discipline.** Hot paths are pre-allocated; the engineering reviewer will find the same memory-discipline patterns familiar from automotive ECU and embedded RTOS work.

§2 · DEMONSTRATOR STATE TODAY

2 What runs, what has been validated

The demonstrator is a Unity-based simulator that drives the same KS decision and comms code that runs in the live PiDog-class hardware integration. The list below covers capabilities that are demonstrable to a visiting reviewer today, not roadmap items.

EMCON V1.0 (TRANSMITTED-ONLY OPERATOR DISPLAYS)

Under enforced radio silence, every operator-visible value flows through central `GetDisplay*` accessors backed by the transmitted-telemetry view. Validated end-to-end against an independent telepathy / RF-silence sweep on 22 May 2026. State propagation, codecs, and per-agent MAVLink emission gating are all on the transmitted path. This is the defence-grade discipline auditors want to see; in civil mode it is the EMCON OPEN preset and the displays render full mesh chatter on the same code path.

BURST MESH RESILIENT DATALINK

Platform-to-platform state propagation as serialised diffs over a burst-queued mesh. Demonstrably resilient under partial link degradation. Architectural choice deliberately mirrors what a real radio mesh will look like — the wire format is portable; the simulator's "two RF" was a sim artefact, not a hidden data plane. The same multi-hop relay covers civil terrain-occluded and post-disaster comms-denied scenarios.

SWARM COORDINATION & KILL-CHAIN

Distributed kill-chain coordination across multiple platforms, with operator-in-the-loop authority levels and per-agent emission gates. Demonstrable swarm-destroy scenarios with full after-action audit record. The same finite-state machine maps onto a civil response chain — see §4.

PIDOG HARDWARE INTEGRATION

Live integration with PiDog-class quadruped ground platform — audio direction-of-arrival sensing, distributed tactical-map awareness, and operator-grade after-action review. Used as a partner demonstrator. Repo is source of truth; deployment is repo-to-Pi, never edited in-place.

TACTICAL-MAP OPERATOR SURFACE

Multi-display tactical map with ghost / uncertainty rendering for transmitted positions under EMCON, dead-reckoned forward markers, and 32×32 imagery thumbnails. UI follows a consistent "hacker-2010 sci-fi" aesthetic: dark background, neon-accent panels, every tab built through a shared C2UIHelper for visual consistency.

MISSION PLANNER & FIRE-REQUEST GOVERNANCE

Operator-driven mission planning with explicit fire-request governance, evasion-budget tracking, and NFZ (no-fly zone) enforcement. Demo missions cover urban survey, swarm-destroy, BDA (battle damage assessment), and EV / ground-vehicle scenarios.

Independent verification

The codebase is held to an honest-caveat audit discipline. Multi-agent dry-runs (an in-house QA technique that hypothetically traces user data across the full codebase with 5-8 parallel scenarios) routinely surface dozens of edge-case findings per session. Each cycle is re-run against patched code to catch issues introduced by the fixes themselves. The Karneth Strata Technical Response (KARNETH_STRATA_TECHNICAL_RESPONSE) documents the methodology and a representative finding-and-fix pass.

Reviewer takeaway: the demonstrator is not stage-managed. It is the same code that will fly on real hardware, exercised with a QA discipline that the team applies to its own work without being asked to. This is the strongest available signal that the company will not surprise an investor with hidden architectural debt at Series A close.

3 How the simulator de-risks each hardware milestone

The standard failure mode for a defence-tech demonstrator is the "simulator gap": the demo runs beautifully in the sim, then collapses on real hardware because the sim hid the hard parts (real radios, real latency, real noise, real safety constraints). Karneth Strata's simulator is engineered against that failure mode from the start.

HARDWARE MILESTONE	WHAT THE SIMULATOR DE-RISKS	WHAT REAL HARDWARE ADDS
Single-platform autonomy	Mission planning, NFZ enforcement, evasion budget, kill-chain authority levels — all exercised in sim with the same decision code that will run on the real platform.	Real flight dynamics, real sensor noise, real battery / endurance constraints.
Multi-platform swarm	BURST MESH propagation, EMCON-coherent state, transmitted-only telemetry, distributed coordination, operator-grade audit record.	Real radio range, real jamming environment, real link quality variance.
Hardware-in-the-loop integration	The Platform Adapter pattern is exercised live against PiDog-class hardware today.	Per-platform integration burn-in; on-vehicle compute envelope; real safety-cutout integration.
Operator-station ergonomics	Full tactical-map, mission-planner, after-action review surfaces validated against demo missions today.	Field ruggedisation, multi-display power envelope, ITAR-controlled distribution discipline.

The validation methodology is explicit: each hardware milestone has named simulator coverage *and* named gaps. Investor and partner reviewers can see exactly what is being claimed and exactly what is not.

4 The same architecture, two markets — wildfire mission walkthrough

This section evidences the executive-summary claim (§3 of the front tier) that the KS architecture is domain-agnostic. The walkthrough is a wildfire response mission run on the same Decision, Comms, and Operator code that today runs the defence demonstrator. Every code path named below is shared between the two domains; the differences are runtime configuration and payload.

Mission setup

- **Scenario.** Wildfire in mountainous terrain. Cellular comms have failed because a relay tower is inside the fire footprint. The fire incident commander (IC) operates a KS station from a forward base camp.
- **Fleet.** Four drones — denoted A1 to A4 below. Each carries a thermal sensor and a retardant payload. Configuration loaded from the same `vehicle_types.json` profile mechanism the defence demonstrator uses, with a civil-fire payload class added (the payload class definition is engineering work scoped post-pilot — see "what is not yet built", below).
- **Mission preset.** EMCON fleet-floor = `OPEN`. Mission-string set loaded from a civil-response vocabulary set. HITL authority chain configured for civil ICS (incident command system) rather than defence ROE.
- **Authority model.** All payload-release actions require explicit IC authorisation. The same fire-request governance pipeline that gates defence engagements gates civil retardant drops. (HITL is, if anything, more important in civil work because of post-incident liability and audit obligations.)

Phase walkthrough — what is on the blackboard

```
PHASE 1   DETECT
          A1 sweeps a grid-pattern survey route.
          Thermal sensor reads 240 degC on coordinates (47.21, -122.34).
          A1 pushes Contact{ id=hot.001, class=fire,
                             owner=A1, conf=0.62, observedAt=t0 }
          onto the platform-shared blackboard.
          BURST MESH propagates the diff to the other agents and the IC.

PHASE 2   CORROBORATE
          A2 routes via terrain to confirm. Independent thermal read at
          the same coordinates.
          A2 pushes Corroborator{ ref=hot.001, owner=A2,
                                   conf=0.71, observedAt=t1 }
          onto the blackboard.
          Confidence on hot.001 is raised in the joint blackboard view
          (corroboration logic = same code as defence CONTACT corroboration).
          Status: classified as active fire, awaiting auth.

PHASE 3   AUTHORISE
          IC's operator station displays the corroborated hotspot.
          IC clicks AUTHORISE on the fire-request governance dialog.
          The action emits a FireAuth{ target=hot.001, granted=true,
                                        grantedAgent=A3, authority=IC.greg,
                                        authoredAt=t2 } record.
          This is the same code path as a defence weapons-release auth,
          with mission-string presets swapped.

PHASE 4   STRIKE (DROP)
          A3 receives the FireAuth diff. Mission planner routes through
          the NFZs the IC has previously marked over the evacuated village.
          A3 executes payload drop on coordinates.
          A3 pushes Strike{ ref=hot.001, agent=A3, executedAt=t3 } onto
          the blackboard. (Same Strike state-machine as defence.)

PHASE 5   ASSESS (BDA)
          A4 conducts post-drop thermal sweep on the drop coordinates.
          Primary drop confirmed: a 60 m radius around the drop point is
          now below 50 degC.
          However, a residual hotspot at the eastern edge (47.22, -122.33)
          still reads 180 degC.
          A4 pushes BdaReport{ ref=hot.001, outcome=PARTIAL,
                                residual=true, observedAt=t4 } onto the
          blackboard.
          Residual triggers a new Contact{ id=hot.002, ... } and the
          cycle resumes at PHASE 2.
          The mission ends when the BDA reads FIRE_OUT (all residuals
          below 50 degC).
```

```
PHASE 6   RESILIENCE (in-flight)
          At t2.4 A1 ducks behind a ridge for terrain reasons.
          Line-of-sight to base camp is lost.
          A4 has line-of-sight to both A1 and base camp.
          A1's PHASE-2 contact reports relay via A4 over BURST MESH
          multi-hop. (Same protocol that handles defence GPS-denied
          terrain.)
          Operator station never loses the picture; the IC sees the
          mesh-relayed reports with a "via A4" sender annotation.
```

Every named entity above (`Contact`, `Corroborator`, `FireAuth`, `Strike`, `BdaReport`) is a real type in the KS codebase today. Civil mode adds a fire-class entry to the Contact classification enum and a fire-retardant entry to the payload class registry; it does not require new decision-layer code paths.

Mapping table — defence concept to civil concept

KS ARCHITECTURAL CONCEPT	DEFENCE MISSION EXPRESSION	CIVIL (WILDFIRE) MISSION EXPRESSION	CODE CHANGE?
EMCON preset	SILENT / BURST / COVERT / OPEN per doctrine	OPEN (no silence requirement)	None — runtime config
Contact classification	enemy, friendly, neutral, unknown	fire, hazard, person, hazmat	Enum entries added
Corroboration	Independent platform contact on same target	Independent platform read on same hotspot	None — same logic
HITL authority gate	Fire-request governance, ROE	Same gate, civil ICS authority model	None — preset
Strike	Ordnance release on target	Retardant / water drop on hotspot	Payload class added
BDA	Target effect assessment	Post-drop thermal sweep, fire-out confirmation	None — same state-machine
BURST MESH	Combat GPS-denied, terrain occlusion	Post-disaster comms-denied, terrain occlusion	None — identical
NFZ enforcement	Operator-defined no-go airspace	Evacuated-village / populated-area exclusion	None — identical
Audit record	After-action review	Post-incident report for the fire authority	None — identical
Mission strings	Combat vocabulary	Response vocabulary	String set swap
Operator surface	C2 tactical map	Civil ICS tactical map	Tab labels swap

What is not yet built — honest scoping

The architecture supports the wildfire mission as described. To deploy the mission in production rather than demonstrate the architectural fit, the following engineering work is scoped post-pilot.

- **Retardant / water-drop payload modelling.** The current payload registry models ordnance. Adding retardant-dispersal ballistics, water-bombing release dynamics, and smother-foam patterns is a payload-class module per the existing payload-class pattern. Not architecturally novel; the work is calibration and integration.
- **Mission-string de-militarisation switch.** A runtime preset to swap combat-vocabulary mission strings for response-vocabulary mission strings (this is the same mechanism the multi-language operator-UI work — also still scoped — will use). Operator UI tab labels, button text, and dialog wording all read from a single string set; switching the set is a configuration change.
- **Civil regulatory compliance package.** UK CAA / EU EASA BVLOS compliance for civil unmanned operations. Separate compliance track from the defence procurement path (UK MoD / DASA / NATO). The CTO's functional-safety heritage and the team's design-in safety discipline transfer; the specific BVLOS evidence package is its own deliverable.
- **Civil-ICS authority model integration.** The fire-request governance pipeline supports civil ICS authority structurally; the specific authority-role mappings for a customer fire service (or equivalent) are a configuration deliverable per customer.

None of the four items above is an architectural risk. They are scoped integration deliverables — payload class, mission-string switch, civil BVLOS compliance pack, civil-ICS authority mapping. Funding shape is flexible: investors who want an early civil demonstrator can stage some of the Series-A capital against the deliverables; alternatively the work can wait for a signed civil customer to unlock paid-for engineering. Either path uses the same underlying platform investment that the defence pilot already funds, and the architectural payoff is the same.

> **Reviewer takeaway:** the dual-use claim in the executive summary is not a marketing assertion. The same Decision, Comms, and Operator code today demonstrating the defence kill chain runs the wildfire response chain above. Civil deployment is gated on a small, named set of integration deliverables, not on architectural rework. The Series-A reviewer should read this as a single-platform engineering investment with two parallel market-access paths.

Other civil missions — same architecture, named briefly

- **Disaster response.** Post-earthquake / post-hurricane / post-flood search-and-rescue coordination. Comms-denied is structural (infrastructure destroyed); BURST MESH multi-hop relay is exactly the right protocol.
- **First-responder coordination.** Multi-agency (fire + ambulance + police + search teams) incident response. Operator-by-exception model with explicit authority gates per agency.
- **Infrastructure monitoring.** Powerlines, pipelines, rail corridors, ports. Recurring-revenue / subscription model. Continuous patrol with anomaly-triggered escalation up the corroborate-

authorise-engage-assess chain.

- **Industrial site security.** Large mines, port facilities, energy installations. Multi-drone autonomous patrols with HITL authority for escalation actions.

§5 · INDUSTRIAL SCALE-UP PLAN

5 From demonstrator to fielded production

This section anchors on the COO's prior career (covered in the Executive Summary §5) and projects it onto the specific KS industrial profile. Three phases.

PHASE 1 — PILOT-CUSTOMER DEPLOYMENT

First named operator partner running KS on real hardware. Team scaling biased toward integration engineers and certification engineers. Industrial discipline modelled on Bombardier's lean / Kaizen production-system rollout: defined processes from day one, not retrofitted. Civil-pilot customer is a viable parallel-track candidate at this stage (per §4).

PHASE 2 — MULTI-CUSTOMER RAMP

Reference customer anchors the commercial pipeline; additional customers onboarded via the Platform Adapter pattern. Headcount ramp modelled on the Manor Racing precedent — seven-to-250 in twelve months, with sub-supplier value chain for non-core work. The COO has run this ramp before. Dual-market demand (defence pipeline + civil pipeline) is absorbed by the same engineering capacity.

PHASE 3 — PRODUCTION DISCIPLINE

Certification package complete; field validation alongside pilot customers; production cadence stable. ERP / PLM rollout modelled on the COO's prior introduction at Lotus F1 — the biggest process change in fifteen years at that organisation, executed without disrupting race-week cadence.

Capital ask, headcount ramp curve, and use-of-funds breakdown are in the financial model held under NDA (Investor Brief §7 / §9). This section is concerned with the *shape* of the industrialisation, not its commercial-confidential terms.

6 Designed-in compliance, not retrofitted

The regulatory landscape for autonomous defence and civil systems is the largest single source of execution risk for any Series-A in this space. Karneth Strata's posture is to engineer for it from the start, on four fronts.

Functional-safety heritage

The CTO's four-decade career in embedded automotive and electric-vehicle control includes deep experience with ISO 26262 functional-safety methodology (ASIL A through D), MISRA C compliance, and Safety Element out of Context (SEooC) discipline on real-time embedded operating systems. The same reflexes show in the KS C# codebase: pre-allocation discipline, explicit safety clamps (altitude, NFZ, evasion bounds), watchdogs, and emergency-mode bounds that cannot be bypassed without explicit operator authority.

This is not branding. It is what an embedded engineer who has shipped at ASIL-D writes by default. A technical reviewer will recognise it on the first read of any hot-path file.

EMCON, MAVLink, and defence comms discipline

EMCON v1.0 enforcement (see §2) is the explicit defence-grade analogue of the embedded safety-clamp discipline. Per-agent MAVLink emission gating is in place; transmitted-only operator displays are verified end-to-end; the architecture treats radio-silence as a code property, not a procedural one.

Civil BVLOS regulatory track

UK CAA and EU EASA BVLOS (beyond visual line of sight) regimes for civil unmanned operations are mature and operational. The KS civil-domain compliance evidence package is a separate deliverable from the defence-procurement compliance work, but it draws on the same designed-in safety discipline. The civil regulatory track is materially faster than the defence-procurement track, which is part of why the civil deployment path (§4) is plausibly first to revenue.

EU AI Act & national doctrine fit

The Lead-Box methodology (Investor Brief §5) is the company's structural answer to EU AI Act compliance, sovereign data residency, and the national-doctrine constraints that European primes operate under. Companion documents *KS National Doctrine Fit v1.2* and *KS EU AI Act Moat v1.0* set this out in detail.

The compliance discipline is a moat. Defence and regulated buyers cannot procure from suppliers who treat safety and compliance as afterthoughts. The KS team's instinct is the opposite: design in, document, test, audit. That instinct is harder to teach than the engineering itself, and few competing Series-A teams have it.

7 What KS owns and why it is defensible

The IP that Karneth Strata UK Ltd owns falls into four categories. All four were developed in-house by the KS engineering team; chain of title is clean.

COMMAND-AND-DECISION CODEBASE

The full ks_drone simulator and live-integration codebase: Operator, Decision, Comms, and Platform Adapter layers. C# / Unity for the operator station and simulator; native bridges to PX4 / MAVLink and PiDog-class hardware. Architecture and module catalogue documented internally. The domain-agnostic structure (see §4) is itself a defensible architectural asset.

BURST MESH PROTOCOL

The transmitted-diff propagation pattern that lets the architecture survive moving from simulator to real radios without redesign. Wire format, EMCON gating, and resilient-burst discipline are KS-original. Defence and civil applications of the protocol are the same code.

EMCON V1.0 ENFORCEMENT MODEL

The display-level enforcement pattern (operator displays read from transmitted-only views, not source values) is a defence-grade discipline asset. Codified, audited, and demonstrable. Treated as a runtime preset means the same model serves civil-OPEN deployments without code duplication.

LEAD-BOX METHODOLOGY

The sovereign AI-engineering environment described in Investor Brief §5. Capital-efficient, compliance-defensible, and a procurement enabler for ITAR / UK MoD / NATO-aligned buyers. Detail in the *Lead-Box Methodology v1.0* companion document.

Defensibility comes from five sources combined: (1) the four-decade embedded heritage of the engineering team — a tacit asset that does not transfer with a code dump; (2) the architectural discipline of the codebase, which compounds over time and is hard for a from-scratch competitor to replicate; (3) the COO's industrial-scale-up record, which is a separately-rare asset; (4) the sovereign-stack positioning, which closes off the procurement-eligibility path that most Series-A competitors will follow; (5) the domain-agnostic architecture that lets one engineering investment serve defence and civil markets without forking the codebase.

§8 · TECHNICAL ROADMAP

8 Six, twelve, and twenty-four month gates

This is the technical-only view of the roadmap; commercial milestones and contract timing are in the Investor Brief §6. Gate criteria are capability-based, not calendar-based — the dates below assume a Series-A close in the stated window.

WINDOW	CAPABILITY GATE	DEMONSTRABLE EVIDENCE
0 – 6 months	Pilot-customer integration ready. First named platform partner onboarded through the Platform Adapter pattern. EMCON v1.1 (with hardware-radio test article). Civil mission-preset preliminary scoping in parallel.	Live integration sessions with the partner platform. Updated demonstrator screenshots and telemetry traces. Second-platform adapter merged.
6 – 12 months	Multi-platform swarm on mixed hardware. Operator-station hardening for partner-site deployment. Compliance package draft for European defence procurement. Civil deliverables (payload class, mission-string switch, civil BVLOS compliance) sequenced according to investor and customer pull — buildable in-window if an early civil demonstrator is wanted, otherwise unlocked by civil customer engagement.	End-to-end mixed-fleet mission with operator station at partner site. Compliance draft circulated to procurement reviewers.
12 – 24 months	Field validation. Maritime platform extension (funded post-Series-A). Certification work toward European defence-procurement requirements. ERP / PLM rollout completed alongside production-discipline cadence. Civil pilot deployment if civil customer is signed in window.	Field-trial after-action reports. Maritime demonstrator. Certification package version 1.0. Production cadence metrics. Civil mission after-action reports if civil pilot in window.

§9 · TECHNICAL RISK REGISTER

9 What could go wrong, technically

This is the technical-only risk view. Commercial, capital, and procurement-timing risks are covered in the Investor Brief §8 and are deliberately not duplicated here.

SIM-TO-REAL GAP (RESIDUAL)

Risk: hardware behaviour diverges from simulator behaviour in a way the architecture did not anticipate.
Mitigation: the architecture is engineered specifically against this failure mode (§3). Each hardware milestone has named simulator coverage and named gaps. Live PiDog integration already exercises the Platform Adapter pattern on real hardware.

REAL-RADIO MESH BEHAVIOUR

Risk: BURST MESH propagation under real-radio jamming and range constraints is harder than the simulator suggests.
Mitigation: the wire format is portable by design; transmitted-diffs, not shared memory. Hardware-radio test article planned in the 0–6 month gate. Comms-layer swap is a layer-scoped change, not a system rewrite.

PLATFORM-ADAPTER ONBOARDING COST

Risk: each new platform partner imposes more integration cost than the architecture predicts.
Mitigation: the JSON-driven vehicle-profile pattern keeps per-platform code surface small. PiDog and PX4 integrations already exercise the adapter pattern. Integration-engineering headcount is an explicit Series-A use-of-funds line.

CERTIFICATION TIMELINE DRIFT

Risk: European defence-procurement certification takes longer than projected.
Mitigation: the safety-and-compliance discipline is designed in from the start. The CTO's ISO 26262 / MISRA / SEooC heritage matters here in a way that a less-experienced team cannot replicate quickly. Compliance work begins inside the 6–12 month window, not as a Phase-3 surprise. Civil BVLOS track (§4) provides an earlier, lower-friction first compliance package.

KEY-PERSON DEPENDENCY ON CTO

Risk: the CTO's four-decade heritage is irreplaceable in the near term.
Mitigation: the engineering bench includes several engineers who have worked with the CTO across multiple companies and decades; continuity reduces single-point-of-failure exposure. Documentation discipline (the internal corpus) captures architectural decisions explicitly.

COMPUTE ENVELOPE ON SMALL PLATFORMS

Risk: the on-vehicle compute envelope for small unmanned platforms constrains what the Decision layer can do locally.
Mitigation: the architecture splits decision authority between operator station and platform; on-vehicle compute requirements are explicit per Platform Adapter. Embedded-grade allocation discipline keeps the on-vehicle footprint small by default.

SINGLE-MARKET EXPOSURE (MITIGATED)

Risk: a defence-pipeline-only Series-A is exposed to procurement cycles, political risk, and the slow gestation of defence sales.
Mitigation: the dual-use operational layer (§4) opens a parallel civil-revenue path that does not require architectural fork or capital diversion. Civil deployment is faster to first revenue and produces validation evidence that strengthens the defence pipeline.

DUAL-USE SCOPE CREEP (MANAGED)

Risk: the dual-use story tempts the company to chase civil customers prematurely and dilute defence-pipeline focus.

Mitigation: civil-mission build deliverables (payload class, string switch, BVLOS compliance, ICS authority mapping) are explicitly scoped as discrete, board-staged work — sequenced against investor appetite for an early civil demonstrator and/or signed civil customer engagement, not bolted opportunistically onto the defence-pilot plan. Defence remains the scale market and the engineering centre of gravity regardless of the funding shape chosen.

What is not on this list, deliberately: we do not believe the core architectural risk is novel. The hard problems (mixed-platform command, EMCON-coherent operator displays, resilient mesh comms, domain-agnostic mission-preset architecture) are understood and demonstrably built. The risks above are execution risks on a known architecture, not invention risks on an unknown one.

§10 · GLOSSARY

10 Terms used in this document

Defence-tech and dual-use unmanned-systems vocabulary is dense and acronym-heavy. This glossary covers the terms used in this document and in the Investor Brief; it is intended for non-specialist readers as much as for specialists who want to be sure the company uses these terms in their standard sense.

ASIL (A–D)

Automotive Safety Integrity Level. The four hazard-classification tiers under ISO 26262. ASIL-D is the most stringent. The KS engineering reflexes are drawn from ASIL-A through ASIL-D embedded automotive work.

BDA

Battle Damage Assessment. Post-engagement evaluation of effect against a target. Demonstrable in the KS demonstrator as a swarm-destroy scenario with after-action review. In civil mode the same state-machine reads as post-drop fire-out confirmation (see §4).

BURST MESH

The Karneth Strata platform-to-platform datalink protocol. Propagates state as transmitted serialised diffs over a burst-queued mesh; resilient under degraded link conditions; EMCON-aware; portable to real radios. Serves combat GPS-denied and post-disaster comms-denied scenarios identically.

BVLOS

Beyond Visual Line Of Sight. The civil-aviation regulatory regime governing unmanned-aircraft operations outside the operator's direct visual range. UK CAA and EU EASA BVLOS regimes are operational and form the civil-deployment regulatory track for KS (see §6).

C2

Command and Control. The decision and authority layer of a defence system. KS is a C2 software platform. In civil mode the same layer is read as the incident command system (ICS) authority layer.

C4ISR

Command, Control, Communications, Computers, Intelligence, Surveillance, Reconnaissance. The broader operational-systems-of-systems category that KS sits within.

DDS

Data Distribution Service. A real-time pub/sub middleware standard used in defence and autonomous-systems work, including with encrypted pre-shared-key channels.

Domain-agnostic

A software architecture property in which the decision, comms, and operator layers do not encode assumptions about a specific operational domain. KS is domain-agnostic in this sense — defence and civil missions run on the same code with different runtime presets (see §4).

EMCON

Emissions Control. Operational discipline of restricting or silencing radio emissions to avoid detection. KS EMCON v1.0 enforces this at the operator-display level: every shown value reads from a transmitted-only view, not the source. Civil missions configure EMCON to OPEN; the same code paths apply.

HIL

Hardware-in-the-Loop. Integration discipline in which real hardware is exercised against the live software stack rather than a simulator-only path. KS PiDog integration is a HIL exercise.

HITL

Human-in-the-Loop. The discipline of requiring explicit human authority at decision gates that have legal, lethal, or liability consequences. Defence weapons-release and civil payload-release both run through the same HITL gate in the KS architecture.

ICS

Incident Command System. The standardised civil-emergency command-and-control structure (fire, ambulance, police, multi-agency response). KS civil missions configure the authority model for ICS rather than military rules-of-engagement (see §4).

ISO 26262

The international functional-safety standard for road vehicles. The methodology (hazard analysis, ASIL classification, SEooC discipline) is a transferable engineering reflex; the KS team carries it from prior automotive work.

ITAR

International Traffic in Arms Regulations. US export-control regime that constrains how defence-related software and data may be processed and where. ITAR forbids most US-hosted AI inference for customer-touching code, which is why the Lead-Box methodology matters.

Kill chain

The sequence of decisions and authorities involved in engaging a target: find, fix, track, target, engage, assess. KS implements distributed kill-chain coordination with explicit operator-in-the-loop authority levels. The same state-machine, with mission-string presets, expresses a civil response chain.

Lead-Box methodology

The KS sovereign AI-engineering environment. Locally-hosted compute keeps models, weights, prompts, code, mission data, and customer payload data inside the engineering boundary. Procurement enabler for ITAR / UK MoD / NATO-aligned buyers. Detailed in *Lead-Box Methodology v1.0*.

MAVLink

The de-facto open communications protocol for unmanned aerial vehicles. KS implements per-agent emission gating on MAVLink as part of EMCON discipline.

MISRA C

A widely-adopted coding-discipline standard for safety-critical embedded C. Compliance is required for most automotive and many defence embedded projects. The KS team's MISRA fluency from prior automotive work transfers as a general code-discipline reflex.

NFZ

No-Fly Zone. Operator-defined airspace constraints; KS enforces NFZ at the Decision Layer and surfaces violations explicitly to the operator with a stated reason. Civil missions use NFZs to exclude populated areas (e.g. evacuated villages during a fire response).

PiDog

A quadruped robotic ground-platform class used as the KS demonstrator hardware partner. Live HIL integration today, including audio direction-of-arrival sensing.

PX4

An open-source flight-control software stack for unmanned aircraft. One of the platforms exercised through the KS Platform Adapter pattern.

SEooC

Safety Element out of Context. An ISO 26262 design pattern for reusable safety components specified without binding to a single vehicle programme. A transferable engineering discipline from automotive into autonomous-systems work.

Sovereign stack

A defence-software stack whose components — software, AI methodology, hardware integration, industrial production discipline — are all in jurisdictions the buyer can inspect, audit, and regulate. The structural KS positioning, not a marketing line.

Sub-supplier value chain

Industrial pattern in which a small core organisation orchestrates a wide network of specialist suppliers for non-core work. Demonstrably used by the COO at Manor Racing F1 and is the model KS expects to scale through.

Document: KS_Feasibility_Seed_Technical_Pack_v1.1.html — Engineering Evidence (Back Tier)
Front tier: KS_Feasibility_Seed_v1.1.html — Executive Summary
Read alongside: KS_Investor_Brief_v1.1.html
Issued: May 2026 · **Version:** v1.1 (adds Dual-Use Operational Layer wildfire mission walkthrough, §4) ·
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